

DeepSeek V4 Flash · DSpark 三配置因子分解

A: mem 0.85 / chunk 16384 B: mem 0.90 / chunk 32768 C: mem 0.85 / chunk 32768
C vs A 固定 mem=0.85，隔离 chunked-prefill-size 纯效应；C vs B 固定 chunk=32768，隔离 mem-fraction-static 纯效应。

最优组合 A/B/C	全局均值吞吐 C	C 峰值显存/GPU	高波动组 A/B/C
2 / 6 / 7	766.4 tok/s	51.92 GiB	6 / 6 / 3

核心结论：三轮各 30 组、900 请求，Error / Timeout 全为 0 / 0。128K 长上下文的性能回退源自 mem-fraction-static=0.90 而非 32K chunk：固定 chunk=32768 时，仅把 mem 由 0.90 降到 0.85，128K 三档并发平均回升 +11.55%；固定 mem=0.85 时，chunk 由 16K 提到 32K，128K 档仅变化 +1.69%。配置 C 取得最高全局均值吞吐 766.4 tok/s、最少高波动组（3 组），显存比 B 低 1.88 GiB。

吞吐与投机解码对比（蓝底为该场景最优）

场景	A Gen	B Gen	C Gen	C vs A	C vs B	A AR	B AR	C AR	A AL	B AL	C AL
1K/C1	167.78	167.45	209.41	+24.81%	+25.06%	0.397	0.399	0.557	2.986	2.996	3.784
1K/C8	804.63	819.50	777.95	-3.32%	-5.07%	0.533	0.472	0.411	3.666	3.355	3.057
1K/C16	1094.26	1391.11	1256.06	+14.79%	-9.71%	0.399	0.541	0.431	2.996	3.703	3.157
4K/C1	178.90	184.04	180.92	+1.13%	-1.69%	0.455	0.477	0.463	3.273	3.382	3.314
4K/C8	752.38	779.36	798.57	+6.14%	+2.46%	0.454	0.461	0.488	3.269	3.306	3.439
4K/C16	1169.05	1135.76	1217.73	+4.16%	+7.22%	0.443	0.431	0.461	3.216	3.159	3.303
16K/C1	203.64	216.16	199.32	-2.12%	-7.79%	0.572	0.624	0.554	3.860	4.122	3.770
16K/C8	846.94	886.19	867.27	+2.40%	-2.13%	0.553	0.602	0.566	3.760	4.008	3.829
16K/C16	1239.98	1281.29	1242.64	+0.22%	-3.02%	0.587	0.600	0.527	3.934	4.000	3.643
64K/C1	230.01	227.42	218.16	-5.15%	-4.07%	0.763	0.752	0.709	4.818	4.761	4.545
64K/C8	961.25	924.21	921.02	-4.18%	-0.35%	0.790	0.726	0.731	4.944	4.627	4.659
64K/C16	1258.97	1283.10	1284.68	+2.04%	+0.12%	0.735	0.761	0.730	4.672	4.803	4.651
128K/C1	215.29	191.65	218.88	+1.67%	+14.21%	0.798	0.687	0.818	4.991	4.433	5.091
128K/C8	916.97	847.18	942.16	+2.75%	+11.21%	0.805	0.718	0.827	5.020	4.588	5.133
128K/C16	1153.14	1062.61	1160.57	+0.64%	+9.22%	0.766	0.655	0.754	4.830	4.281	4.769

延迟与重复稳定性

场景	A TTFT	B TTFT	C TTFT	A TPOT	B TPOT	C TPOT	A 波动	B 波动	C 波动
1K/C1	208.0	207.0	206.4	5.764	5.775	4.578	3.54%	0.05%	1.04%
1K/C8	587.3	381.8	460.9	8.648	8.643	9.146	11.03%	6.12%	4.78%
1K/C16	1883.5	1448.5	936.4	11.841	9.272	10.773	14.42%	19.73%	1.14%
4K/C1	210.0	208.1	206.2	5.395	5.243	5.365	6.05%	7.39%	15.79%
4K/C8	540.6	428.0	300.6	9.061	9.020	8.706	8.16%	6.78%	1.87%
4K/C16	1211.0	1222.3	949.9	11.168	11.415	10.626	4.31%	8.99%	2.54%
16K/C1	233.6	234.0	234.3	4.701	4.402	4.794	10.59%	1.16%	3.20%
16K/C8	493.1	498.4	361.0	8.144	7.645	7.881	0.05%	0.31%	3.25%
16K/C16	1596.4	1285.6	932.2	8.991	9.045	9.316	1.86%	0.16%	3.19%
64K/C1	344.1	345.2	344.2	4.019	4.065	4.251	6.04%	3.82%	0.79%
64K/C8	772.9	796.4	728.6	6.914	7.144	6.946	0.10%	4.44%	10.23%
64K/C16	2188.7	2154.6	2277.7	8.480	7.880	8.399	0.17%	16.06%	7.70%
128K/C1	487.8	493.1	488.4	4.172	4.740	4.095	2.75%	1.67%	3.06%
128K/C8	1075.1	1067.4	1071.9	7.093	7.702	6.923	1.27%	0.26%	2.29%
128K/C16	3395.6	3462.6	3360.2	8.197	9.586	8.502	2.88%	2.03%	1.45%

汇总统计

指标	A	B	C
全局均值吞吐 tok/s	746.21	759.80	766.36
峰值吞吐 tok/s	1260.03	1528.32	1334.13
单流 C1 均值 tok/s	199.12	197.34	205.34
Accept Rate 全局均值	0.6033	0.5936	0.6019
Accept Len 全局均值	4.0157	3.9683	4.0097
单流 C1 Accept Rate	0.5971	0.5877	0.6202
单流 C1 Accept Len	3.9856	3.9387	4.1009
峰值显存 / GPU (GiB)	48.58	53.80	51.92
高波动组数 (>5%)	6	6	3
成功 / 错误 / 超时	900/0/0	900/0/0	900/0/0

GPU0-7 跨组资源均值

GPU	A Util	B Util	C Util	A Mem	B Mem	C Mem	A Peak	B Peak	C Peak	A W	B W	C W
GPU0	95.7%	95.6%	98.0%	46.97	50.19	48.26	48.40	53.61	51.73	185.1	184.5	186.5
GPU1	96.2%	96.6%	97.2%	47.16	50.37	48.45	48.58	53.80	51.92	190.6	189.7	191.6
GPU2	96.3%	96.3%	97.4%	47.16	50.37	48.45	48.58	53.80	51.92	192.5	191.8	193.1
GPU3	95.9%	95.5%	97.1%	47.16	50.37	48.45	48.58	53.80	51.92	187.8	187.9	189.3
GPU4	95.6%	95.7%	96.0%	47.16	50.37	48.45	48.58	53.80	51.92	183.0	182.6	183.9
GPU5	96.1%	95.7%	97.2%	47.16	50.37	48.45	48.58	53.80	51.92	189.4	189.0	191.4
GPU6	94.6%	95.3%	96.9%	47.16	50.37	48.45	48.58	53.80	51.92	191.0	190.8	191.3
GPU7	93.1%	94.5%	96.3%	46.97	50.19	48.26	48.40	53.61	51.73	183.8	184.3	185.2

配置 A: mem 0.85 / chunk 16384 • 30 组详细结果

Ctx	C	R	Gen	AR	AL	TTFT mean/p50/p95	TPOT	ITL	Peak GiB	Power	OK	Err/TO
1K	1	1	170.75	0.408	3.041	208.0/207.4/211.8	5.659	5.659	45.39	161.0	30	0/0
1K	1	2	164.81	0.386	2.931	208.0/207.6/211.3	5.870	5.870	45.39	161.6	30	0/0
1K	8	1	760.25	0.545	3.726	789.3/228.6/2577.2	8.887	8.887	45.39	180.7	30	0/0
1K	8	2	849.02	0.522	3.606	385.3/230.2/1306.8	8.409	8.409	45.39	196.4	30	0/0
1K	16	1	1015.33	0.400	3.003	2826.4/3432.0/4382.1	11.920	11.920	45.76	212.0	30	0/0
1K	16	2	1173.18	0.398	2.988	940.6/462.1/1958.9	11.761	11.761	45.76	221.8	30	0/0
4K	1	1	173.49	0.434	3.170	209.3/209.4/212.3	5.565	5.565	45.76	161.3	30	0/0
4K	1	2	184.32	0.475	3.375	210.6/209.4/215.5	5.225	5.225	45.76	161.4	30	0/0
4K	8	1	721.69	0.442	3.211	806.8/236.0/2153.2	9.189	9.189	45.76	188.2	30	0/0
4K	8	2	783.07	0.465	3.327	274.4/228.5/402.1	8.934	8.934	45.76	192.3	30	0/0
4K	16	1	1143.85	0.452	3.258	1488.0/1795.2/1861.4	10.944	10.944	45.76	214.4	30	0/0
4K	16	2	1194.25	0.435	3.174	934.0/545.3/1608.1	11.392	11.391	45.77	220.6	30	0/0
16K	1	1	192.85	0.528	3.637	233.6/232.1/247.0	4.962	4.962	47.50	160.8	30	0/0
16K	1	2	214.43	0.617	4.083	233.6/232.2/241.5	4.439	4.439	47.50	160.4	30	0/0
16K	8	1	846.74	0.566	3.824	611.2/265.4/1524.2	8.049	8.049	47.50	196.7	30	0/0
16K	8	2	847.13	0.539	3.696	375.0/262.2/646.9	8.240	8.240	47.50	196.2	30	0/0
16K	16	1	1228.44	0.584	3.919	2013.2/3187.9/3216.6	9.225	9.225	47.51	203.7	30	0/0
16K	16	2	1251.51	0.589	3.949	1179.6/790.4/3224.7	8.756	8.756	47.51	211.6	30	0/0
64K	1	1	223.05	0.732	4.664	347.0/339.1/402.2	4.148	4.148	48.44	157.6	30	0/0
64K	1	2	236.96	0.794	4.972	341.1/336.1/351.9	3.890	3.890	48.44	157.6	30	0/0
64K	8	1	961.73	0.790	4.942	862.6/537.8/1879.7	6.946	6.946	48.45	196.4	30	0/0
64K	8	2	960.76	0.790	4.947	683.3/394.5/1412.6	6.883	6.883	48.45	196.1	30	0/0
64K	16	1	1257.90	0.747	4.733	2405.8/2457.1/6942.2	8.395	8.395	48.46	208.7	30	0/0
64K	16	2	1260.03	0.723	4.611	1971.6/2449.7/2460.4	8.566	8.566	48.46	209.0	30	0/0
128K	1	1	218.25	0.815	5.074	487.3/486.5/501.5	4.109	4.109	48.57	152.9	30	0/0
128K	1	2	212.33	0.782	4.909	488.2/483.3/514.2	4.236	4.236	48.57	152.0	30	0/0
128K	8	1	922.77	0.818	5.087	1118.4/689.5/2282.0	6.918	6.918	48.58	193.9	30	0/0
128K	8	2	911.17	0.791	4.954	1031.7/571.1/2270.1	7.267	7.267	48.58	196.0	30	0/0
128K	16	1	1169.77	0.801	5.007	3388.1/4051.8/7540.2	7.974	7.974	48.58	206.7	30	0/0
128K	16	2	1136.51	0.731	4.654	3403.0/4183.4/7338.4	8.421	8.421	48.58	209.8	30	0/0

配置 B: mem 0.90 / chunk 32768 • 30 组详细结果

Ctx	C	R	Gen	AR	AL	TTFT mean/p50/p95	TPOT	ITL	Peak GiB	Power	OK	Err/TO
1K	1	1	167.41	0.399	2.996	206.1/206.1/208.2	5.777	5.777	47.26	160.8	30	0/0
1K	1	2	167.49	0.399	2.995	207.9/206.9/216.0	5.773	5.773	47.26	161.2	30	0/0
1K	8	1	794.42	0.463	3.312	405.7/227.3/1165.1	8.659	8.659	47.26	189.3	30	0/0
1K	8	2	844.57	0.480	3.398	357.8/226.4/1030.9	8.626	8.626	47.26	192.2	30	0/0
1K	16	1	1253.90	0.538	3.686	2243.3/2938.5/2944.8	9.553	9.553	47.69	204.1	30	0/0
1K	16	2	1528.32	0.543	3.721	653.7/470.2/905.8	8.991	8.991	47.69	214.4	30	0/0
4K	1	1	177.23	0.451	3.251	207.6/206.7/210.6	5.445	5.444	47.69	160.9	30	0/0
4K	1	2	190.84	0.502	3.512	208.5/208.0/214.6	5.041	5.041	47.69	161.2	30	0/0
4K	8	1	752.93	0.448	3.245	584.8/228.4/1528.2	9.161	9.161	47.70	197.8	30	0/0
4K	8	2	805.80	0.473	3.366	271.3/226.9/399.6	8.878	8.878	47.70	195.2	30	0/0
4K	16	1	1084.69	0.443	3.220	1579.9/1650.1/2396.1	11.412	11.412	47.70	209.0	30	0/0
4K	16	2	1186.83	0.419	3.098	864.7/543.7/1770.8	11.419	11.419	47.70	223.9	30	0/0
16K	1	1	214.90	0.620	4.099	233.4/231.1/249.5	4.429	4.429	49.38	160.8	30	0/0
16K	1	2	217.41	0.629	4.146	234.6/232.9/243.7	4.374	4.374	49.38	159.4	30	0/0
16K	8	1	887.57	0.618	4.089	605.4/262.9/1512.6	7.497	7.497	49.38	193.5	30	0/0
16K	8	2	884.80	0.586	3.927	391.5/332.5/597.5	7.794	7.794	49.38	194.1	30	0/0
16K	16	1	1280.25	0.610	4.053	1637.0/2252.5/2259.2	9.078	9.077	49.38	206.6	30	0/0
16K	16	2	1282.33	0.589	3.948	934.3/775.0/1337.5	9.012	9.012	49.38	216.0	30	0/0
64K	1	1	231.76	0.772	4.860	343.7/338.5/359.3	3.982	3.982	53.60	157.3	30	0/0
64K	1	2	223.09	0.732	4.661	346.7/340.6/392.2	4.147	4.147	53.60	156.4	30	0/0
64K	8	1	944.75	0.766	4.826	831.4/543.2/1892.6	6.977	6.977	53.61	191.4	30	0/0
64K	8	2	903.68	0.686	4.429	761.3/642.1/1420.1	7.312	7.312	53.61	196.2	30	0/0
64K	16	1	1386.12	0.869	5.347	2070.1/2441.8/4636.1	7.652	7.652	53.61	202.6	30	0/0
64K	16	2	1180.08	0.653	4.259	2239.1/2276.3/7395.5	8.108	8.108	53.61	200.5	30	0/0
128K	1	1	193.25	0.695	4.475	493.7/487.8/531.0	4.696	4.696	53.79	153.8	30	0/0
128K	1	2	190.05	0.679	4.392	492.4/487.1/526.5	4.784	4.784	53.79	153.9	30	0/0
128K	8	1	848.28	0.707	4.536	1101.2/593.8/2410.7	7.742	7.742	53.79	198.4	30	0/0
128K	8	2	846.08	0.728	4.639	1033.7/569.3/2285.9	7.661	7.661	53.79	195.3	30	0/0
128K	16	1	1073.38	0.675	4.376	3473.7/3795.5/8244.3	9.458	9.458	53.80	209.1	30	0/0
128K	16	2	1051.84	0.636	4.185	3451.4/3908.3/7780.0	9.714	9.714	53.80	211.5	30	0/0

配置 C：mem 0.85 / chunk 32768 • 30 组详细结果

Ctx	C	R	Gen	AR	AL	TTFT mean/p50/p95	TPOT	ITL	Peak GiB	Power	OK	Err/TO
1K	1	1	208.32	0.553	3.766	205.5/205.4/208.1	4.604	4.604	45.39	161.9	30	0/0
1K	1	2	210.50	0.561	3.803	207.2/206.0/213.3	4.552	4.552	45.39	162.7	30	0/0
1K	8	1	759.35	0.420	3.103	645.2/392.2/1894.1	9.123	9.123	45.51	190.5	30	0/0
1K	8	2	796.55	0.403	3.011	276.5/226.4/386.6	9.169	9.169	45.51	193.8	30	0/0
1K	16	1	1248.90	0.427	3.135	986.7/627.3/2727.6	10.762	10.762	45.53	217.8	30	0/0
1K	16	2	1263.22	0.436	3.179	886.1/459.5/2642.2	10.784	10.784	45.53	222.1	30	0/0
4K	1	1	166.63	0.408	3.041	207.1/207.7/212.6	5.804	5.804	45.61	161.4	30	0/0
4K	1	2	195.20	0.517	3.588	205.4/204.8/208.0	4.927	4.927	45.61	161.5	30	0/0
4K	8	1	791.09	0.479	3.392	323.2/230.0/523.5	8.741	8.741	45.61	196.2	30	0/0
4K	8	2	806.05	0.497	3.486	278.0/227.1/406.1	8.672	8.672	45.61	194.2	30	0/0
4K	16	1	1202.28	0.458	3.287	1096.5/616.4/2536.8	10.643	10.643	45.66	221.1	30	0/0
4K	16	2	1233.18	0.463	3.318	803.4/544.3/1208.2	10.608	10.608	45.66	222.9	30	0/0
16K	1	1	202.51	0.568	3.836	235.9/232.8/255.6	4.712	4.712	47.50	160.6	30	0/0
16K	1	2	196.13	0.541	3.704	232.7/230.9/242.1	4.876	4.876	47.50	159.2	30	0/0
16K	8	1	853.16	0.526	3.628	357.0/259.9/628.4	8.353	8.353	47.50	199.8	30	0/0
16K	8	2	881.38	0.606	4.031	365.0/260.4/584.7	7.409	7.409	47.50	194.9	30	0/0
16K	16	1	1222.85	0.513	3.574	930.0/883.3/2075.2	9.839	9.839	47.51	215.1	30	0/0
16K	16	2	1262.44	0.542	3.712	934.5/781.5/2388.7	8.792	8.792	47.51	212.6	30	0/0
64K	1	1	217.30	0.705	4.524	343.5/338.6/370.2	4.270	4.270	51.72	155.8	30	0/0
64K	1	2	219.02	0.713	4.566	344.8/336.8/388.4	4.232	4.232	51.72	154.9	30	0/0
64K	8	1	873.93	0.699	4.497	721.8/389.9/1458.5	6.953	6.953	51.73	190.2	30	0/0
64K	8	2	968.12	0.764	4.822	735.3/550.5/1391.6	6.939	6.939	51.73	194.0	30	0/0
64K	16	1	1334.13	0.772	4.857	2416.1/2316.5/7163.0	8.152	8.152	51.74	208.9	30	0/0
64K	16	2	1235.23	0.689	4.446	2139.2/2436.7/2638.2	8.646	8.646	51.74	207.5	30	0/0
128K	1	1	222.23	0.834	5.171	489.4/479.7/534.9	4.024	4.024	51.91	151.5	30	0/0
128K	1	2	215.54	0.802	5.011	487.5/484.3/512.6	4.166	4.166	51.91	151.3	30	0/0
128K	8	1	952.95	0.859	5.295	1069.4/599.8/2302.2	6.599	6.599	51.91	192.6	30	0/0
128K	8	2	931.37	0.794	4.971	1074.4/570.8/2295.9	7.247	7.247	51.91	194.9	30	0/0
128K	16	1	1169.00	0.773	4.861	3376.8/4477.6/7172.7	8.390	8.389	51.92	211.2	30	0/0
128K	16	2	1152.14	0.736	4.677	3343.6/3912.5/7463.5	8.615	8.614	51.92	209.5	30	0/0

结论与测量口径

因子分解表明：chunked-prefill-size 由 16384 提升到 32768 在中短上下文与高并发场景带来正向收益，且不是 128K 回退的原因；mem-fraction-static 由 0.85 提升到 0.90 会压缩可用 KV cache 余量，在 128K 长上下文下显著降低 Accept Rate 与吞吐。因此推荐配置 C（mem 0.85 + chunk 32768）：全场景均衡最佳、重复稳定性最好、显存占用低于 B。
Gen 为组级聚合吞吐；AR/AL 来自对应组的服务日志字节区间；TTFT/TPOT/ITL 来自流式响应；GPU 指标每 0.5 秒采样；波动为两次完整重复的吞吐相对差异，>5% 标红提示需追加验证。